P-P-553a

P-P-553a
August 5, 1963

SUPERSEDING
Int. Fed. Spec. P-P-00553 (Army-GL)
December 11, 1962,
Fed. Spec. P-P-554
September 25, 1956, and
Fed. Spec. P-P-552a
August 12, 1953

FEDERAL SPECIFICATION

POLISH, FURNITURE, LIQUID

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers liquid polishes for cleaning and polishing wood and metal furniture (see 6.1).

1.2 Classification.

1.2.1 *Grade, types, and classes.* The polishes covered by this specification shall be of one grade and shall be of the following types and classes, as specified (see 6.2):

Type I.—Liquid emulsion cleaner and polish with oil.

Type II.—Liquid emulsion cleaner and polish.

Class 1.—Wax emulsion.

Class 2.—Wax emulsion with silicone.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-C-96—Cans, Metal; 28 Gage and Lighter.

PPP-C-186 — Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 7930

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society For Testing And Materials Standards (ASTM):

D 94-62—Saponification Number by Color-Indicator Titration

D 1289-55T—Nonvolatile Matter (Total Solids) in Water-Emulsion Waxes (Tentative).

D 92-57—Flash and Fire Points by Cleveland Open Cup.

D 86-62—Distillation of Petroleum Products.

E 70-52T—Determination of pH of Aqueous Solutions with the Glass Electrode (Tentative).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample approval. When specified (see 6.2), before production is commenced, a one-gallon sample of the finished item shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.1.3. The approval of the preproduction sample authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with all applicable provisions of the specification. The preproduction sample shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production item.

3.2 Material.

3.2.1 *Type I.* The liquid emulsion cleaner and polish with oil shall consist of a stable emulsion of oil in water and shall be free from abrasives and dyes.

3.2.2 *Type II, class* 1. The liquid emulsion cleaner and polish with wax shall be a stable system of wax and volatile aliphatic hydrocarbons, suitably emulsified in water. The emulsion shall be a free-flowing liquid which can be easily applied and spread with a cotton cloth. The polish-cleaner shall remove dirt and tarnish and shall impart a glossy lustre to wood or metal furniture when applied as directed.

3.2.3 *Type II, class* 2. The liquid polish shall be a stable system of wax, silicone oil and volatile aliphatic hydrocarbons, suitably emulsified in water. The emulsion shall be a free-flowing liquid which can be easily applied and spread with a cotton cloth. The polish-cleaner shall remove dirt and tarnish and shall impart a glossy lustre to wood or metal furniture when applied as directed.

3.3 Chemical and physical requirements. The polishes shall conform to the requirements in table I when tested as specified in table II.

3.4 Stability properties. The polishes shall show no evidence of settling or breakdown of the emulsions when tested as specified in 4.4.12.

3.5 Application. The polishes shall flow freely and spread easily on the surface to which it is applied when tested as specified in 4.4.13.

3.6 Gloss increase.

3.6.1 *Type I.* The gloss of the panel after polishing shall increase not less than 25 percent when tested as specified in 4.4.14.

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TABLE I.—*Chemical and physical requirements*

Property	Type I		Type II—Class 1		Type II—Class 2	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Oil content, percent	40.0	—	—	—	—	—
Oil and water content, percent	98.0	—	—	—	—	—
Free caustic alkali	—	None	—	None	—	None
Saponification number	—	15	—	—	—	—
Non volatile matter, percent	—	—	4.0	8.0	6.0	12.0
Silicone oil, percent	—	—	—	—	2.0	4.0
Volatile hydrocarbon, percent	—	—	20.0	40.0	20.0	40.0
Flash point, °F.	—	—	95	—	95	—
Distillation range °F.	—	—	284	430	284	430
pH value	—	—	6.0	9.0	6.0	9.0
Abrasiveness	—	None	—	None	—	None

3.6.2 *Type II, class* 1. The gloss of the polished panel shall increase not less than 25 percent when tested as specified in 4.4.14.

3.6.3 *Type II, class* 2. The gloss of the polished panel shall increase not less than 40 percent when tested as specified in 4.4.14.

3.7 **Gloss retention.** The gloss of the polished panel shall decrease not more than 5 percent when tested as specified in 4.4.14.

3.8 **Odor.** The polishes shall not have an objectionable odor.

3.9 **Workmanship.** The end item shall be clean, homogeneous and free flowing, and shall conform to the levels of quality established herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.1.2 *Inspection of components and materials.* In accordance with 4.1 the supplier shall be responsible for insuring that the quality assurance provisions for components or materials shall be in accordance with this specification and with referenced subsidiary specifications to the extent applicable. In the event of conflict, this specification shall govern.

4.1.3 *Preproduction sample inspection.* When required, the preproduction sample submitted in accordance with 3.1, shall be inspected as specified in 4.2.3 and 4.2.4.

4.2 **Inspection of the end item.**

4.2.1 *Examination of the end item.* The end item shall be examined for the defects in the applicable sub-paragraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.6. The lot size for determining the sample size in accordance with MIL-STD-105, shall be expressed as indicated in the applicable examination paragraph.

4.2.2 *Examination of the end item for defects in odor and appearance.* The sample units for this examination shall be the contents of one filled container. The lot size for

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this examination shall be expressed in limits of filled containers.

Examine	Defect
Odor	Rancid or objectionable odor.
Type	Not type specified.
Condition	Not free from abrasives, added colors or dyes.
	Not free flowing liquid.

4.2.3 *Examination of the end item container for defects in appearance design and construction.* The sample unit for this examination shall be one filled container. The lot size for this examination shall be expressed in units of filled containers.

Examine	Defect
Container, glass	Any cord, blister, or stone.
	Any crack or thin spot.
Closure	Not metal or plastic continuous thread screw cap.
	Cap not equipped with non-absorbent lining material.
	Lining material not resilient in itself or backed by a resilient material.
	Lining material does not remain in cap upon removal of cap.
	Cap not tightened to prevent leakage.
Can	Not double seamed and cemented.
	Cement soluble in polish.
	Body not one piece.
	Body side seam not soldered.
	Excess solder.
Can closure (gallons only)	Not screw neck with cap.
	Cap not lined to prevent leaks.

4.2.4 *Examination of the end item for defects in average net content.* The sample unit for this examination shall be one filled container. The lot size shall be expressed in units of filled containers. The lot shall be unacceptable if the average net content for all filled containers examined is less than the specified or indicated quantity.

4.2.5 *Examination of preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, contents, and weights comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Packaging	Not as specified.
	Unit containers not packaged as specified (see section 5).
Packing	Not as specified.
	Unit containers not snugly packed.
	Damaged container, affecting serviceability.
	Container material not as specified.
	Closures not accompanied by specified or required methods or materials.
	Number of unit containers less than specified or indicated.
Marking	Illegible, incomplete, not as specified, or not in accordance with contract requirements.

4.2.6 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels (for determining the sample size) and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

	Inspection levels	*AQLs*
For examination in 4.2.2 ..	I	2.5
For examination in 4.2.3 ..	I	2.5
For examination in 4.2.4 ..	L-4	—
For examination in 4.2.5 ..	L-4	4.0

4.3 **Testing of the polish.** The polish shall be tested for the applicable characteristics as indicated in table II, from each lot presented for acceptance. The sample unit shall be one quart of the end item. The sample size shall be 5. Each sample unit shall be taken from a different shipping container of the lot. The test sample shall consist of a

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TABLE II.—*Instructions for testing end item (composite)*

Characteristic	Specification reference		Number determinations composite	Results reported as	
	Requirement	Test method		Pass or fail[1]	Numerically to nearest[2]
Oil	Table I	4.4.1	2	—	0.1 percent
Water and oil content (type I)	Table I	4.4.2	2	—	0.1 percent
Free caustic alkali	Table I	4.4.3	1	X	—
Saponification number (type I)	Table I	4.4.4	2		1.0 number
Nonvolatile matter (type II)	Table I	4.4.5	2		0.1 percent
Silicone oil content (type II, class 2)	Table I	4.4.6	2		0.1 percent
Volatile hydrocarbon content (type II)	Table I	4.4.7	2		0.1 percent
Flash point (type II)	Table I	4.4.8	2		1.0°F.
Distillation range (type II)	Table I	4.4.9	2		1.0°F.
pH value	Table I	4.4.10	2		0.1 pH
Abrasive properties	Table I	4.4.11	1	X	—
Stability properties	3.4	4.4.12	1	X	—
Application	3.5	4.4.13	1	X	—
Gloss increase	3.6	4.4.14	1 (average of 10 readings)		1.0 percent
Gloss retention	3.7	4.4.14	1 (average of 10 readings)		1.0 percent

[1] If failure is indicated, report either description of failure or numerical point of failure as applicable.
[2] Test reports shall include all values on which results are based.

2-quart composite of the 5 sample units. For characteristics where results are reported as variables the average of the two determinations shall comply with the requirements. For characteristics where results are reported as attributes, all determinations shall comply with the requirements.

4.4 Test Procedures.

4.4.1 *Oil (type I only).* Place approximately 100 grams of the sample, accurately weighed, in a beaker and dilute to about 200 ml. with distilled water. Heat mixture and acidify with 0.5*N* hydrochloric acid, using methyl orange as indicator. After oil has collected at the top, carefully add 50- to 100-ml. of benzene, transfer to a separatory funnel and wash the benzene solution of the oil with water until free of acids. Draw off the water layer and transfer the benzene layer to a weighed beaker. Evaporate the benzene on the steam bath and dry the oil in an oven at 100° -105°C. for 24 hours. Cool in a desiccator and weigh. Calculate the percentage of oil in the sample.

4.4.2 *Water (type I only).* Distill 100 grams of the sample, using a distillation flask connected to a condenser and having 100-ml. graduated cylinder for a receiver. The initial boiling point shall be not lower than 100°C. Stop the distillation at 100°C. The distillate represents the water or volatile part of the polish.

4.4.2.1 *Calculation:*

$$\text{Water, percent} = \frac{V \times 0.998}{S} \times 100$$

where:

V = volume of water in milliliters at 20°C.
S = weight of sample in grams.

4.4.3 *Free caustic alkali.* A 10-ml. sample of the polish shall be digested in 100-ml. of neutral ethyl alcohol (94 percent or higher). Add a few drops of phenolphthalein indicator and if no pink color results at this point no free caustic alkali is present. Should a pink color appear, dilute about 10 ml. of an original sample of polish with 10 ml. of distilled water, boil and test the vapors for

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ammonia. If present, continue boiling until the ammonia is driven off, cool, add 100 ml. of neutral ethyl alcohol (94 percent or higher). Add a few drops of phenolphthalein indicator. No pink discoloration shall result.

4.4.4 *Saponification number (type I).* The saponification number (type I) shall be determined in accordance with ASTM Method D 94-62.

4.4.5 *Nonvolatile matter (type II).* The percentage nonvolatile matter (total solids) shall be determined in accordance with ASTM Method D 1289-55T.

4.4.6 *Silicone oil content (dimethyl-siloxane polymer) (type II, class 2 only).* Accurately weigh 3 to 4 grams of polish into a tared platinum or vycor crucible. Place in an oven maintained at 105° ± 2.5°C. for 2 hours to remove volatile matter and then cool in a desiccator. To reduce subsequent frothing of sample, add 4 or 5 drops of chlorobenzene and agitate or warm contents to effect solution. Add 2 milliliters of fuming sulfuric acid (15 percent) to the solution at room temperature. Add 0.5 milliliter of fuming nitric acid immediately following the addition of the sulfuric acid. Heat crucible and contents over a low flame for several hours to accomplish digestion. When the mass has solidified and danger of frothing is past, heat the crucible over a Meker burner to complete expulsion of sulfur trioxide fumes. Place residue in a muffle furnace for 1 hour at 800°C. Cool in a desiccator and weight residue.

4.4.6.1 *Calculation.* Calculate percent silicone oil as follows:

$$\text{Percent silicon dioxide} = \frac{\text{weight of residue}}{\text{weight of sample}} \times 100$$

$$\text{Percent dimethyl siloxane polymer (silicone oils)} = \text{percent silicon dioxide} \times \frac{74}{60}$$

4.4.7 *Volatile petroleum hydrocarbon content (type II only).* To 500 grams of the furniture polish in a 1-liter round bottom flask, add sufficient dilute hydrochloric acid to break the emulsion. Add boiling chips to prevent bumping and connect the flask to a water cooled condenser whose outlet is joined to a 500-ml. flask by a 105° vented connector tube. Ground glass joints are desirable. Heat carefully to distill over the volatile material using a suitable hot plate, electric mantle or oil bath. Separate the petroleum hydrocarbons in the distillate from the water with a separatory funnel and transfer it to a tared beaker. Determine the weight of the petroleum hydrocarbon and compute the percentage. Save the separated hydrocarbon for the determinations in 4.4.8 and 4.4.9.

4.4.8 *Flash point of the volatile petroleum hydrocarbon.* The flash point shall be determined in accordance with ASTM Method D 92-57.

4.4.9 *Distillation range of volatile petroleum hydrocarbon.* The distillation range shall be determined in accordance with ASTM Method D 86-62.

4.4.10 *pH of emulsion (type II).* The pH shall be determined in accordance with ASTM Method E 70-52T, using the glass electrode.

4.4.11 *Abrasive properties.* Spread a thin film of polish over a microscope slide. Place another microscope slide over the film of polish. Press the two slides together using a slight pressure and at the same time sliding one glass slide over the other. The movement between the glass slides shall be smooth without any indication of the presence of abrasive.

4.4.12 *Stability properties.* A graduated glass cylinder about 1 inch in diameter shall be filled with a sample of the polish to a height of 10 inches. The cylinder shall be glass-stoppered and left undisturbed at a temperature of 20° to 25°C. for 48 hours. The polish shall remain as a uniform emulsion without settling (layer formation) and

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without breakdown (appearance of oil) of the emulsion.

4.4.13 *Application.* Carry out test under condition of 20° to 30°C. Pour a small amount of polish onto a soft cloth. The polish shall flow free from the container. Apply the polish with the cloth to a finished wood and metal surface. The polish shall spread easily to a thin film. Rub the film of polish with a dry cloth, using no more than moderate pressure. The polished surface shall be free from oiliness or tack and shall have a good lustre.

4.4.14 *Gloss increase and retention.* Select a clean rubbed-varnished or rubbed-lacquered panel approximately 1 by 2 feet, having a gloss meter reading between 20 and 30, using a Gardner 60° Glossmeter. A minimum of ten readings shall be taken to determine an average, moving and rotating the glossmeter 90° after each reading so that the entire surface shall be represented by the average. This average is the gloss reading of the panel. Apply the polish as specified in 4.4.11, and directly after buffing, determine the gloss reading. The percentage increase in gloss shall be not less than 25 percent for type I and type II, class 1 and not less than 40 percent for type II, class 2. Store the panel for 24 hours at 77°F. and 50 percent ± 2 percent relative humidity and again obtain the average gloss reading to determine gloss retention. The gloss of the polish shall not decrease more than 5 percent.

5. PREPARATION FOR DELIVERY

For civil agency procurement, Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Levels A and B.*

5.1.1.1 *Type I.* The type I furniture polish shall be packaged in one-pint, one-quart or one-gallon (see 6.2) metal, glass or plastic cans or bottles conforming to type V, class 4, exterior coating plan B of PPP-C-96; group A, class 1, grade 2, style 1, closure A or B or group A, class 2, style 1, closure A or B of PPP-C-186, respectively. The one-pint cans and plastic bottles of polish shall be packaged in accordance with the interior packaging requirements in the Appendix of PPP-C-96. The glass bottles of polish shall be packaged in accordance with PPP-C-186.

5.1.1.2 *Type II.* The type II furniture polish shall be packaged in one-quart glass bottles conforming to group A, class 1, style 1, closure A or B of PPP-C-186. The bottles of polish shall then be packaged in accordance with PPP-C-186.

5.1.2 *Level C.* The furniture polish shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The furniture polish, packaged in metal cans or plastic bottles as specified in 5.1, shall be packed in accordance with the overseas shipment requirements for filled cans in the Appendix of PPP-C-96. The furniture polish of one class only, packaged in glass bottles as specified in 5.1, shall be packed in accordance with the level A requirements of PPP-C-186.

5.2.2 *Level B.* The furniture polish, packaged in metal cans or plastic bottles as specified in 5.1, shall be packed in accordance with the domestic shipment requirements for filled cans in the Appendix of PPP-C-96. The furniture polish of one class only, packaged in glass bottles as specified in 5.1, shall be packaged in accordance with the level B requirements of PPP-C-186.

5.2.3 *Level C.* The furniture polish, packaged as specified in 5.1, shall be packed

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in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The furniture polishes covered by this specification are intended primarily for the cleaning and polishing of furniture and other finished wood and metal surfaces. The polishes are intended for varnished, lacquered, or enameled furniture finishes. They are not intended for exterior or floor surfaces.

6.2 **Ordering data.** Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type required (see 1.2).
(*c*) Whether or not a preproduction sample is required (see 3.1).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(*e*) Quantity of polish desired (see 5.1.1.1).

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Compound, polishing, not otherwise indexed by name, furniture,
Carload minimum weight 36,000 pounds.

Motor:
Compounds, polishing, not otherwise indexed, furniture.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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Copies of this specification may be purchased for cents each.

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